

Transformer differential protection using wavelet transform



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ABSTRACT

This paper will propose a cascade of minimum description length criterion with entropy approach along with artificial neural network (ANN) as an optimal feature extraction and selection tool for a wavelet packet transform based transformer differential protection. The proposed protection method provides a reliable and computationally efficient tool for distinguishing between internal faults and inrush currents. The role of minimum description length criterion with entropy approach has been found to improve the efficiency of ANN with the dimensionality reduction of the feature vector. This reduction plays a major role in preventing the redundancy effect that can occur when using several features in an intelligent based monitoring system.

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1. Introduction

Power transformer protection is a priority for power utilities because power transformers are among the most expensive and reliable types of equipment in the power system network. Transformer differential protection allows only for internal fault and proposed scheme provides stability against external faults, presenting a good dynamic and steady-state performance, as it can be observed in the simulation and real time results. However, the reliability of the differential relay may be compromised due to malfunctioning when an inrush current phenomenon occurs. Therefore, differential relays should be equipped with a discrimination facility to detect inrush currents as a normal transient state and internal faults' currents.

In order to overcome some of the concerns that have been raised in the use of the second restraint method [1], several intelligent based monitoring techniques have been alternatively recommended for differential current classification. Artificial neural networks (ANN) were proposed for discriminating transformer inrush currents from internal fault currents. Both time response [2] and consecutive windowed samples were successful features implemented in the ANN based methods [3,4]. Probabilistic neural networks (PNN) with particle swarm optimization for attaining optimal smoothing factor was another approach proposed for enhanced efficiency of the classifier. The proposed feature vector

was composed of voltage to frequency ratio along with the amplitude of the current signal [5].

Wavelet based feature extraction of transformer differential current was proposed as an enhanced classification rate with different approaches [6,7]. The difference between the amplitudes of wavelet coefficients was proposed in order to obtain faster performance and reduce the dependency on certain thresholds in the relaying scheme [6]. A cascade of artificial neural networks and wavelet transform has proven to be efficient in obtaining acceptable discrimination accuracy. The feature vector is determined by the composition of the energies of the first, second and third detail signals that are decomposed from the discrete wavelet transform of the differential current signal. The detail spectral energies of level-3 decomposition were determined at three different instances, forming a 9-feature-input vector to the ANN [8]. Another proposed method for pattern recognition of the differential current signal is a combination of decision trees and wavelet analysis. In addition to wavelet coefficients, other features, such as the RMS and percent differential currents, restraining current, second and fifth harmonics, were applied [9]. More stable features were generated for differential current monitoring by extracting fewer coefficients for internal fault and inrush current detection [10]. Wavelet packet transform with optimal selection of the mother wavelet and resolution level, which are implemented using the minimum description length criterion (MDL), have been successfully applied for transformer differential protection [11]. Unlike [11], this paper proposes not only optimal mother wavelet family but also presents optimal decomposition level. Moreover, the proposed paper uses ANN approach to discriminate inrush and

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internal fault conditions thanks to extensive computer simulations. Additionally, CT saturation concept with Jiles–Atherton modeling technique is also taken into account in the simulations. The efficiency of the wavelet packet transform was also verified with a novel implementation using the Butterworth passive filters [12].

It is also worth mentioning that all of the proposed techniques have shown an outstanding classification rate, but with less emphasis on the feature selection step for optimal classification. Although MDL has been implemented in [11] for optimal selection of the mother wavelet and resolution level and the optimal number of wavelet coefficients were proposed in [9], less effort was necessary to optimize the feature vector for dimensionality reduction. In a smart grid environment where reliability is a crucial aspect, the redundancy effect of the feature vector of an intelligent based monitoring system must be prevented. Feature selection for optimal combination of features is an essential step to reduce the redundancy effect. This redundancy may affect the overall performance of the monitoring system by reducing the classification capability [13].

This paper presents a new scheme for power transformers differential protection in which the concept of the optimized neuro-wavelet technique is introduced. This includes a novel selection algorithm of wavelet based transformer differential current. The MDL and the entropy criterion are used for optimal selection of the mother wavelet and the resolution level and present dimensionality reduction in the feature vector. The energy levels of the detail coefficients are used as feature vector. The validity of the proposed algorithm is tested through ANN based classifier of power transformer inrush and internal fault differential currents as well as differential currents resulted from terminal faults. The suggested algorithm enlightens the idea of an optimal selection technique as opposed to an arbitrarily selection technique, which ensures an optimal selection of differential current features in the ANN classifier. The proposed technique has achieved the maximum possible classification rate with the minimum number of features used to monitor the transformer inrush and internal fault currents.

2. Materials and methods

2.1. Wavelet based feature extraction using MDL and entropy criteria

Real time classification of power transients is very challenging because the high frequency content superimposed on the power frequency signals are usually non-periodic, short term, and non-stationary waveforms. Wavelet transform (WT) is proposed in order to extract discriminative features which will help in differentiating between transients associated with islanding event and those created from any other event, such as temporary fault. Similar to Fourier transform, which breaks the signal into sinusoidal waves of different frequencies, WT breaks the signal into a shifted and dilated version of a short term waveform called the mother wavelet. Mathematically, the continuous wavelet transform (CWT) of a signal can be represented in (1).

$$CWT_x^{(\varphi)}(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t) \varphi^* \left(\frac{t-b}{a} \right) dt \quad (1)$$

where a is the scale, b is the translation or position, $i_d(t)$ is the analyzed differential current signal, and φ is the mother wavelet. The definition of CWT demonstrates how the wavelet analysis is a measure of the resemblance between the wavelet and the original signal. The calculated coefficient refers to the correlation or similarity between the function and the wavelet at the current scale. If the coefficient is relatively large then the signal is similar to the wavelet at this point in the time-scale plane. In the practical

implementation of CWT there will be redundant information; therefore, for the ease of computational purposes, the scale and translation variables are discretized. The discrete wavelet transform is described in (2)

$$DWT_{\varphi} f(m, n) = \sum_k f(k) \varphi_{m,n}^*(k) \quad (2)$$

where, $\varphi_{m,n}$ is the discretized mother wavelet. In wavelets applications, choosing an appropriate mother wavelet plays an important role in analysis and depends on the application. Various basis functions have been proposed, including Haar, Morlet, Mexican, Daubechies, bi-orthogonal, etc. While selecting the most suitable wavelet family in analyzing power system transients is of vital importance, the selection of the optimal wavelet family is essential for optimal performance of the classifier. Therefore, discrete wavelet transform (DWT) is used with MDL and entropy criteria for optimal selection of the mother wavelet and the resolution level. DWT basically decomposes the differential current signal into approximation and details different resolution levels which can be employed for detecting the transient nature of the transformer inrush and internal fault currents. The differential current signal is decomposed when the sampling frequency is 4 kHz and the window length is one cycle. Standard deviation is used to represent the energy of the nominated details as input features [15]. It should be noted that the proposed algorithm is employed to improve the selection of the wavelet-based transformer differential current features, thereby improving the classifier performance, rather than developing the wavelet based classifier itself.

MDL data criterion is used for the selection of a wavelet function and applied to the simulation and real time data. The Shannon entropy based criterion is used in order to find an optimum level of resolution for the proposed protection algorithm. The entropy based criterion calculates the entropy of each subspace consisting of detail coefficients ('d') and approximation coefficients ('a') at each level of resolution of the DWT. It compares the entropy of a parent subspace with those of its children's subspaces in order to find out the optimum level of resolution using the optimum mother wavelet. The criterion states that if the entropy of a signal at a new level is higher than that of the previous level, the decomposition of the signal is not needed [16]. A similar dimension reduction method is given in [17] to distinguish two classes, which is based on descriptive statistics i.e. order 1 and 2 conditional moments of the predictors. The proposed method does not require any additional calculation and present high accuracy.

2.2. Artificial neural network classifier

Feed forward multi-layer perceptron ANN with back propagation algorithm is used as a tool to evaluate the proposed optimal feature vector and compare its efficiency with other commonly used features. Fig. 1 shows a general structure for ANN used in pattern classification applications. where details (d_1 , d_2 , and d_3) in the first layer are sent to the hidden layer in weighted links. The net activation of the hidden layer is calculated as in (3)

$$net_j = \sum_{i=1}^d d_i w_{ji} + w_{j0} \quad (3)$$

where 'd' is the number of details recommended in the feature modeling process and ' w_{ij} ' are the weights between the 'ith' input unit and the 'jth' hidden unit. The output of the hidden layer is a nonlinear function of its net activation as shown in (4)

$$y_j = f(net_j) \quad (4)$$

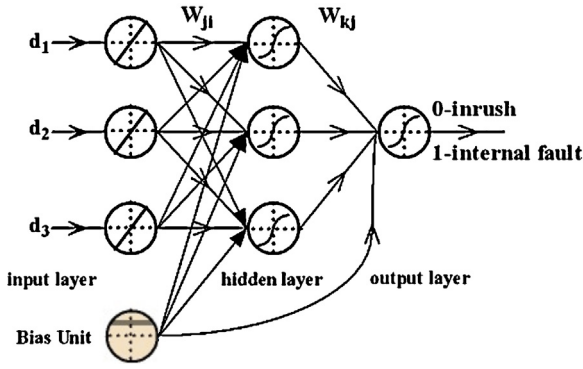


Fig. 1. Structure of artificial neural network used evaluation of the feature vector.

where ' y_i ' is the output of the hidden layer. The net activation of the output layer is determined as shown in (5)

$$net_k = \sum_{i=1}^{N_h} y_i w_{ki} + w_{k0} \quad (5)$$

where ' N_h ' is the number of hidden neurons and ' w_{kj} ' is the weights between the ' k 'th input unit and the ' j 'th hidden unit. The output of the output layer is calculated as a nonlinear function of its net activation as shown in (6)

$$z_j = f(net_k) \quad (6)$$

where ' z_k ' is the output at the ' k 'th output unit. In this study two classes (outputs) are possible: 1 for internal fault class and zero for normal operation class (inrush current). Continuous tan sigmoid activation function, as expressed in (7), is used as the activation function.

$$f(net_{j,k}) = \frac{2}{1 + e^{-2net_{j,k}}} - 1 \quad (7)$$

The resulting discrimination function of the neural network at the end of its output layer can be expressed as per (8).

$$g_k(x) = z_k = f \left(\sum_{i=1}^{N_h} w_{kij} f \left(\sum_{i=1}^d d_i w_{ji} + w_{j0} \right) + w_{k0} \right) \quad (8)$$

The inputs (d_1 , d_2 , and d_3) of ANN are the detail coefficients of the differential current obtained through CTs. Differential current has 80 samples in a period. A rectangular window is used for all calculations and that has to have at least 6 decomposition level in wavelet calculation ($\log_2(80) \approx 6.32$). However, only 3-decomposition-level is required to get the maximum performance by means of suggested selection algorithm.

The proposed feature vector is evaluated and compared with other commonly employed combinations details. These combinations are decomposed using bior3.3, bior3.5, bior3.7, and bior4.4 at resolution levels between two to seven. Six iterations are conducted for each feature vector tested in the ANN. Round Robin testing based algorithm, where leave-1-out strategy is implemented, is used to enhance the statistical significance of the ANN based evaluation. The algorithm is cycled over all the available feature samples where all the samples are used for training except the one left for testing. Then the previously tested sample is returned to the training vector and another unused sample is taken for testing, given that it has not been used before for testing. The process is repeated until all the possible feature samples are tested.

Table 1
Cases simulated in PSCAD®.

Simulations	Runs
Internal fault current	78 (portion of turns shorted are: 5%, 15%, 25%, 40%, and 50%)
Inrush current (residual flux effect)	104 (8 different voltage levels of the residual flux source 10–80% of the rated main voltage source)
Sympathetic inrush current	13
No-load inrush current	13
Inrush due to external fault clearance	20

2.3. Simulation

Simulation is performed on a delta-wye transformer with ratings of 315 MVA and 400/220 kV. PSCAD/EMTDC software is used to generate training and test signals under different operating conditions of the transformer that are normal, over-excitation, magnetizing inrush, sympathetic inrush, and internal fault [14]. In the high side, there is a source 200 MVA, 400 kV with 10Ω as internal impedance; in low side, there is a three-phase load of 285 MW and 137.28 MVAR. Different situations are simulated and summarized in Table 1.

The first case demonstrates inrush current signal generation when the transformer is energized at no load. Thirteen no-load inrush current signals are generated by adjusting the energization breaker's switching angle from 0° to 360° in increments of 30° . The second case simulated is the sympathetic inrush. Similarly, 13 sympathetic inrush current signals are generated by increasing the breaker's energization switching angle from 0° to 360° in increments of 30° . For an inrush due to residual flux, a parallel voltage source is connected to the main source through a breaker. The two sources are 180° apart and operate at two distinct periods, where the residual-flux voltage source will inject the current before energization. The value of the injected current reflects the value of the residual flux in the transformer core prior to energizing the transformer. Therefore, the magnitude of the residual flux can be controlled by controlling the operational instant of the flux source. To obtain all the possible values of the residual flux in the transformer core, it is necessary to adjust the angles of the breaker at values between 0° and 360° in increments of 30° and adjust the operating voltage of the flux source at values between 10% and 80% of the rated voltage. Inrush current simulation due to external fault clearance is also taken into account. To simulate this condition load is switched off at different switching angles between 0° and 190° .

The internal fault of the power transformer is simulated using the special transformer internal fault module available in the PSCAD® library. The transformer turns were shorted at different values such as 5%, 15% and 80% of the total transformer turns through a breaker. Different internal fault signals are generated by shutting off the breaker at different switching angles between 0° and 360° of the voltage source cycle.

Fig. 2 shows the simulation environment for all possible working conditions of power transformers.

As seen in Fig. 2, primary and secondary currents are obtained through CTs. A non-linear model based on Jiles-Atherton modeling is used to model the CTs under saturation conditions. Therefore, proposed protection technique takes into account CT saturation on both sides of the transformer. Secondary current of CT has higher order harmonics both even and odd ones and among them the 2nd and 5th harmonic components are dominant during saturation. Traditionally, harmonic restraint approach is employed to detect the ratio of magnitudes, i.e. 2nd/1st and 5th/1st. However, it is not reliable due to several aspects and the differential protection

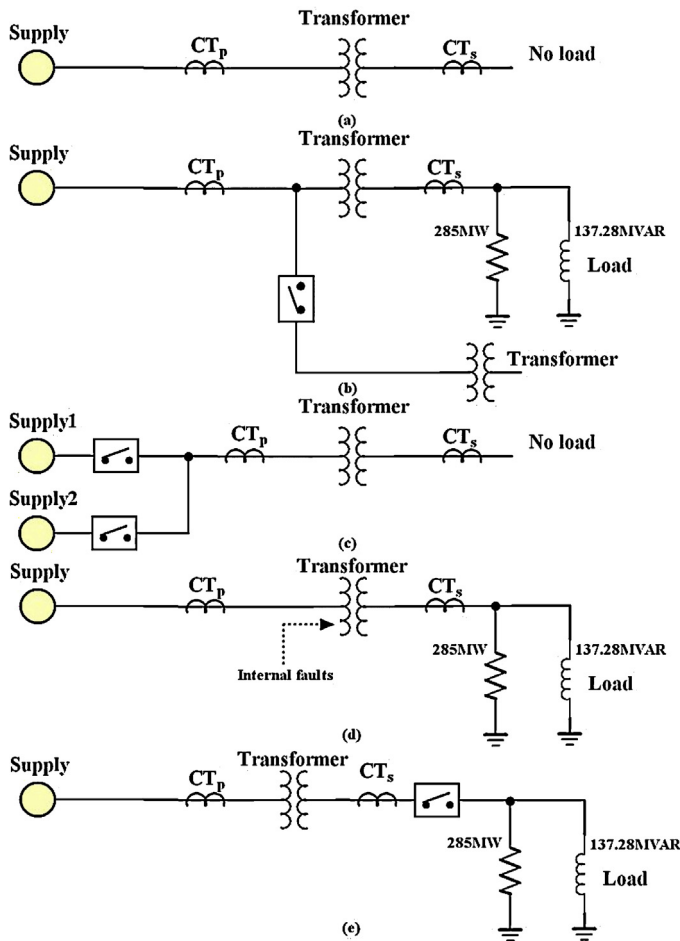


Fig. 2. Simulation case studies: (a) no-load inrush current, (b) sympathetic inrush current, (c) the effect of residual flux in core, (d) internal fault simulation, and (e) inrush current due to clearance of external fault.

concept requires more intelligent methods to discriminate inrush and internal fault conditions. During CT saturation the resulted harmonics will affect the energy levels of the associated detail coefficients (according to their frequency ranges) and this solution

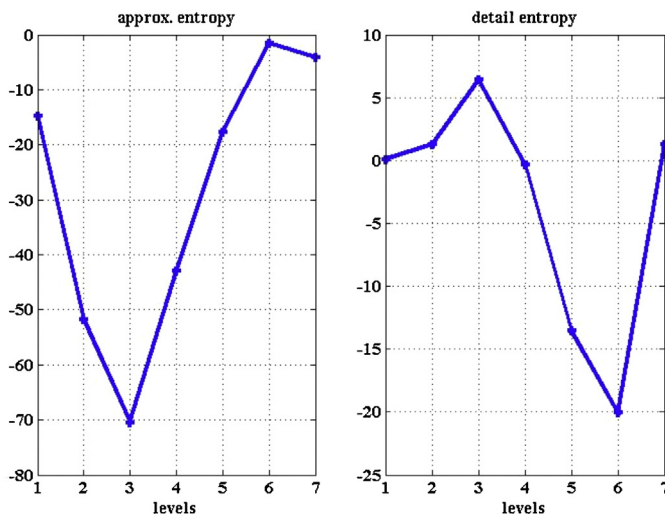


Fig. 3. Calculation of the optimal decomposition level (inrush current decomposition).

Table 2

MDL results for a test signal.

Filter	MDL	Filter	MDL	Filter	MDL	Filter	MDL
db1	2.62	sym2	1.25	coif1	1.80	rbio1.1	2.62
db2	1.25	sym3	2.60	coif2	4.81	rbio1.3	3.81
db3	2.60	sym4	5.37	coif3	5.59	rbio1.5	2.62
db4	2.90	sym5	5.15	coif4	5.40	rbio2.4	3.57
db5	3.98	sym6	1.54	coif5	5.39	rbio2.6	5.16
db6	5.24	sym7	5.65	bior1.1	2.62	rbio2.8	3.57
db7	0.42	sym8	4.94	bior1.3	0.55	rbio3.3	2.78
db8	1.53	sym9	1.69	bior1.5	1.08	rbio3.5	1.74
db9	3.49	sym10	1.44	bior2.4	0.59	rbio3.7	2.78
db10	3.22	sym11	3.61	bior2.6	2.30	rbio4.4	4.82
db11	4.28	sym12	3.73	bior2.8	2.30	rbio5.5	0.38
db12	3.91	sym13	0.50	bior3.3	0.31	rbio6.8	3.97
db13	4.07	sym14	4.09	bior3.5	0.41		
db14	3.39	sym15	3.33	bior3.7	4.84		
db15	2.61	sym16	4.22	bior4.4	1.43		
db16	3.42	sym17	1.99	bior5.5	1.87		
db17	5.49	sym18	3.67	bior6.8	4.84		
db18	4.65	sym19	4.98				
db19	4.20	sym20	0.94				
db20	2.93						

db: daubechies, sym: symlets, coif: coiflets, bior: bi-orthogonal, and rbio: reverse bi-orthogonal wavelet filters.

The bold values are optimal values for each wavelet family.

may require higher decomposition levels for detail coefficients. The suggested protection technique also proposes an optimal decomposition level taking into account of CT saturation.

3. Results and discussions

MDL and Shannon entropy techniques are applied to the simulation data (150 inrush and 78 internal fault) employing the most common 68 mother wavelet families. The optimal wavelet filters are demonstrated in Table 2 among 68 mother wavelet families.

Db7, sym13, coif1, bior3.3, and rbio5.5 wavelet filters are found to be the best for any given same test signal. The mathematical backgrounds of wavelet filters can be found in Ref. [18]. Since bior 3.3 has the minimum coefficient, it is used for further analysis in DWT. The optimal decomposition level is found to be 3 according to the Shannon entropy (Fig. 3). Bior3.3 coefficients recommended by MDL, particularly at the 3rd resolution level, have optimal correlation (acceptable correlation with minimum dimensionality possible) than the analyzed mother wavelets, as shown in Table 2 and Fig. 3. Although such results have been clearly obtained in this paper, all in all, feature selection algorithms are highly sensitive to the signals employed and therefore transformer

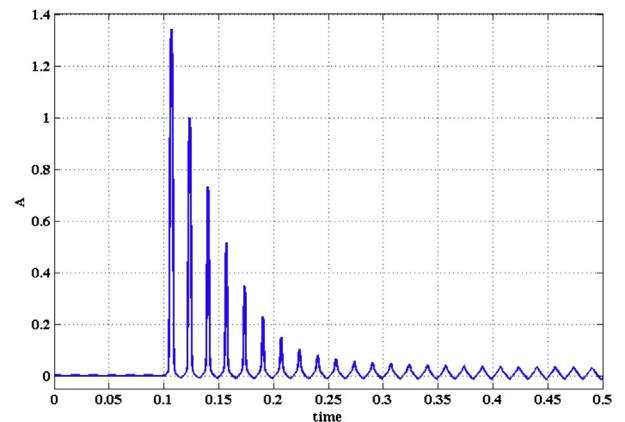


Fig. 4. A sample inrush current simulation.

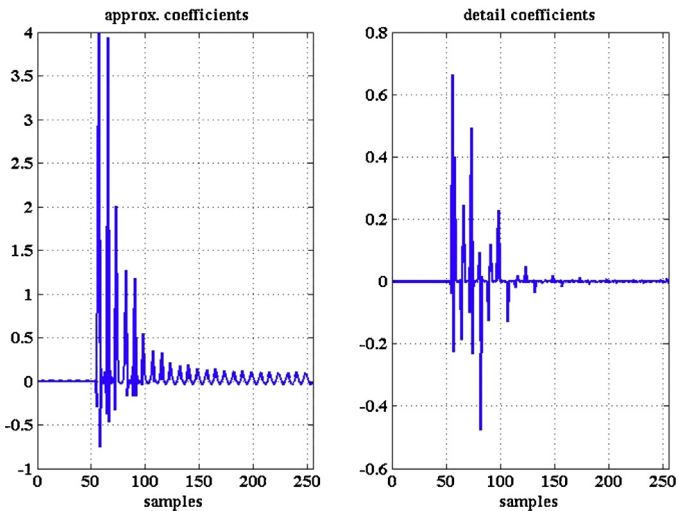


Fig. 5. Decomposition of the sample inrush current using the optimal wavelet family at the 3rd level.

conditions, such as transformer loading, fault type, inrush type, etc.

In Fig. 3, an inrush current is tested. The analyzed signal is given in Fig. 4 and its wavelet decomposition using the optimal wavelet family (bior3.3) at the 3rd level is given in Fig. 5.

The selected optimal wavelet function, *bior3.3*, is also tested by the different wavelet families with the levels from 1 to 7. Fig. 6 shows the classification rate (with 95% confidence interval) for the neuro-wavelet based detection method of the transformer inrush and internal fault differential currents. It is evident that at wavelet resolution levels between 3 and 4, the classification rate starts to saturate for the optimal mother wavelet, *bior3.3*. The proposed MDL and entropy approaches can be, therefore, verified for optimal selection of the resolution level. However, compared to other commonly used mother wavelets from the same family, no significant difference can be detected for the classifier performance when the proposed ‘*bior3.3*’ is used. It seems that the classification efficiency is highly dependent on the time-frequency characteristics of the transient signal in general with less dependence on the type of basis function, i.e. mother wavelet.

The classification accuracy of 100% was obtained by using 50% training and 50% testing data. The efficiency of the proposed technique was also tested using different percentages of training and testing data with the optimal wavelet and decomposition level (*bior3.3* and level 3). Fig. 7 demonstrates the efficiency of the proposed protection algorithm.

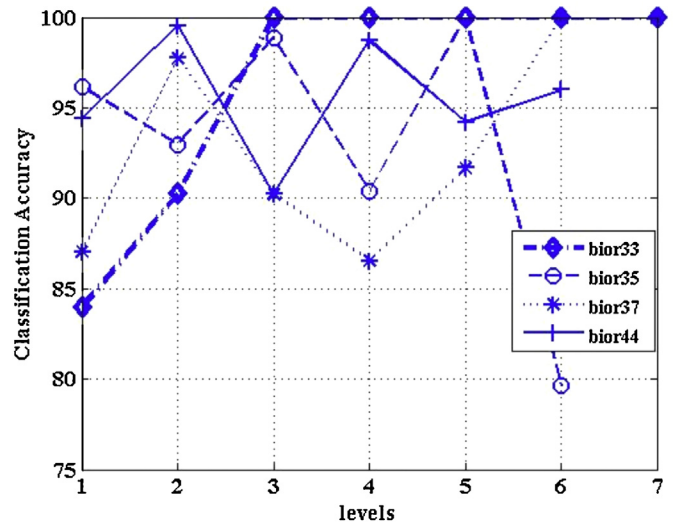


Fig. 6. The dependence of the classification rate of the neuro-wavelet based differential protection on the mother wavelet type and the resolution level.

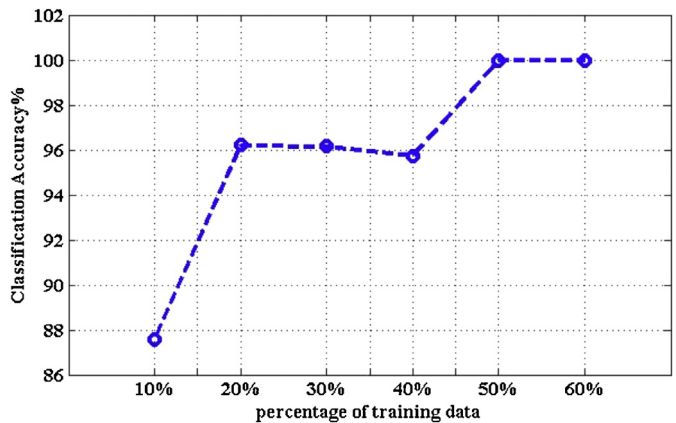


Fig. 7. Testing the protection algorithm with different percentages of training data.

As seen from Fig. 7, the suggested protection algorithm with the optimal mother wavelet and decomposition level has high classification accuracy, even when using low percentage training data, i.e. 96.20% accuracy with 20% training data.

The wavelet based feature extraction and selection processes for the offline training of the ANN is summarized in Fig. 8.

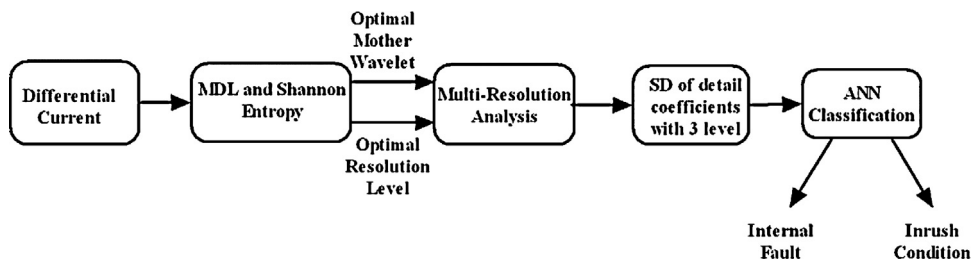


Fig. 8. Feature extraction and selection processes for the offline training of the ANN for the neuro-wavelet based transformer differential inrush and internal fault current detection.

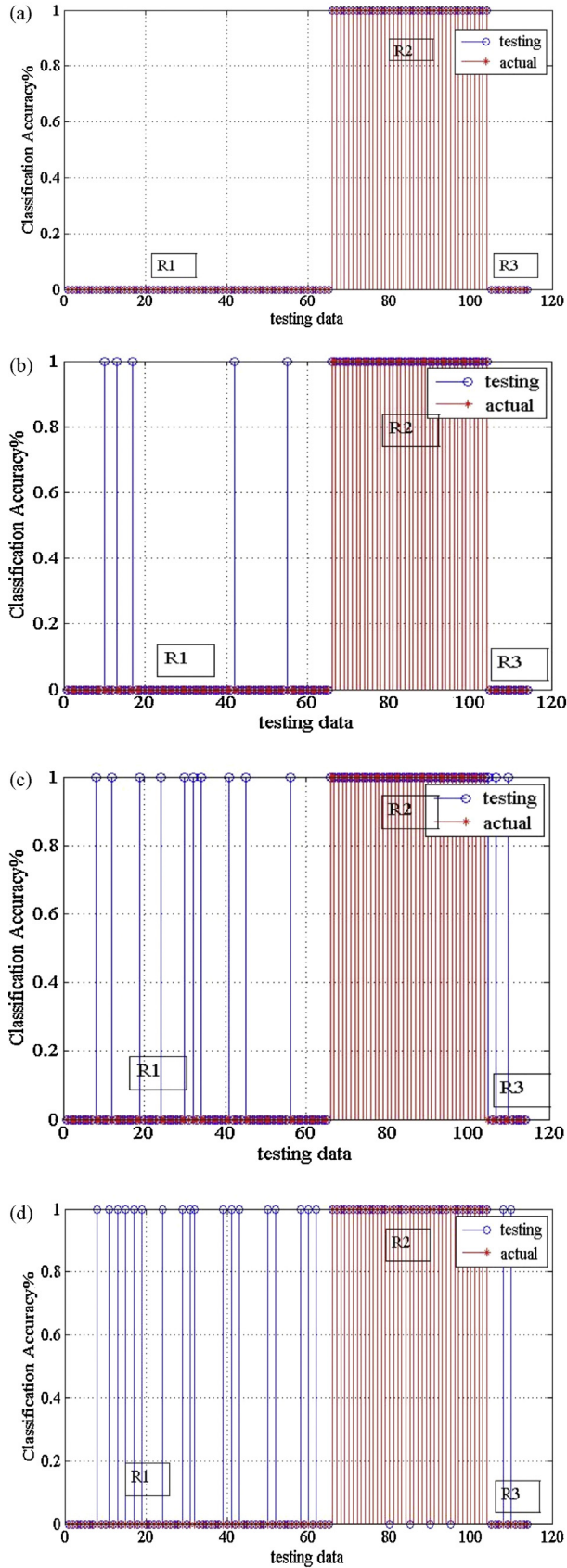


Fig. 9. (a) Performance of optimal wavelet function (*bior3.3*) on testing data (CA%=100). (b) Performance of *bior3.5* wavelet function on testing data (CA%=95.61). (c) Performance of *bior4.4* wavelet function on testing data (CA%=88.59). (d) Performance of *bior3.7* wavelet function on testing data (CA%=78.94).

Table 3

Number of terminal faults.

Terminal faults	Number of experiments on H.V. side	Number of experiments on L.V. side
Phase to phase	10	10
Phase to earth	10	10
Three-phase	10	10

H.V.: high voltage, L.V.: low voltage.

Fig. 9 shows the classification accuracy (in percentage) of the suggested protection algorithm. In addition to optimal wavelet function, *bior3.5*, *bior3.7*, and *bior4.4* wavelet filters are also tested with the decomposition level of 3. For 104 training and testing scenarios, the training time is approximately 43 ms and testing time is approximately 8.3 ms for these trials. Each testing scenario takes approximately 0.0798 ms for ANN classification plus 0.25 ms for acquisition time of each sample. The total testing time is then approximately 0.3298 ms, which is quite an acceptable time interval to distinguish internal and inrush conditions.

In Fig. 9a–d, the region R1 represents inrush fault conditions, R2 represents internal fault conditions, and R3 represents inrush fault conditions due to clearance of an external fault.

3.1. Terminal faults as internal faults

Apart from internal winding faults, terminal faults are performed at the terminals of the transformer. Both the terminals as well as the winding faults are serious in nature. These faults include phase to phase fault on H.V. or L.V. terminals, phase to earth faults on H.V. or L.V. terminals, and three-phase faults on H.V. or L.V. terminals. Additionally, these faults are performed when the transformer is in its rated load. Faults inception angles from 0°C to 90°C are taken into consideration for terminal faults.

A total of 60 experiments were done and 20% of experiments was used as training while the 80% ones was used as testing trials. In these trials, optimal wavelet function (*bior 3.3*) was used. The fault duration was set to one period and a fault resistance was used for simulating high impedance faults between the H.V. and L.V. terminals. Table 3 shows the number of terminal faults.

The experimental setup is shown in Fig. 10.

Since terminal faults produce higher magnitudes than that of winding faults, the proposed wavelet based protection algorithm has successfully identified all the terminal faults with a 100% accuracy. As an example, Fig. 11 shows the performance evaluation of three-phase terminal fault both in H.V. and L.V. side of the transformer.

As stated in Section 1, traditionally harmonic restraint and blocking algorithms based on Discrete Fourier Transform are used for differential relaying purposes. However, they present severe drawbacks and security in practical applications. Fig. 12 shows the typical inrush current and its associated 2nd and 5th harmonic components with respect to time. This technique is prone to misoperation in the presence of transformer inrush currents, which result from transients in transformer magnetic flux. The solution is to introduce an intentional time delay, desensitizing the relay for a given time, adding a voltage signal to restrain, or supervising the differential relay. These solutions need an additional time and reduce the relay response. A more detail comparison is given in Table 4.

4. Conclusions

This paper presents a novel approach for the differential protection of power transformers. Optimal selection of feature vector for a neuro-wavelet based detection of transformer differential inrush and internal fault currents has been successfully performed using a cascade of ANN with entropy approach. Selection of the optimal mother wavelet is found to be the primary need in applying MDL and entropy approach for the selection of the resolution level and the combination of details used. This paper has highlighted the importance of using feature selection methods in implementing an intelligent based monitoring of power system equipment for reduction of the classifier dimensionality, and thus redundancy.

Many case studies including terminal faults were used in testing the proposed algorithm. The simulation results showed fast, accurate, discriminative, and reliable capabilities of the proposed algorithm to identify different natures of currents flowing in a power transformer under different operating conditions. The proposed MDL and Shannon entropy methods for optimizing the wavelet filters ensures a robust performance under inrush currents and provides high sensitivity for the detection of internal faults with a quite acceptable time burden. Furthermore, the suggested algorithm considers CT saturation and changes in power system configuration.

The proposed method can be implemented as a stand-alone differential protection scheme or can be used as an improvement to existing protection systems.

Further research concerning the refinement of the proposed protection technique dealing with real time application is currently in progress.

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